

Thirteenth International Olympiad, 1971

1971/1.

Prove that the following assertion is true for $n = 3$ and $n = 5$, and that it is false for every other natural number $n > 2$:

If $a_1, a_2, \dots, a_n$ are arbitrary real numbers, then

$$(a_1 - a_2)(a_1 - a_3) \cdots (a_1 - a_n) + (a_2 - a_1)(a_2 - a_3) \cdots (a_2 - a_n) \\ + \cdots + (a_n - a_1)(a_n - a_2) \cdots (a_n - a_{n-1}) \geq 0$$

1971/2.

Consider a convex polyhedron P_1 with nine vertices $A_1, A_2, \dots, A_9$; let P_i be the polyhedron obtained from P_1 by a translation that moves vertex A_1 to A_i ($i = 2, 3, \dots, 9$). Prove that at least two of the polyhedra $P_1, P_2, \dots, P_9$ have an interior point in common.

1971/3.

Prove that the set of integers of the form $2^k - 3$ ($k = 2, 3, \dots$) contains an infinite subset in which every two members are relatively prime.

1971/4.

All the faces of tetrahedron $ABCD$ are acute-angled triangles. We consider all closed polygonal paths of the form $XYZTX$ defined as follows: X is a point on edge AB distinct from A and B ; similarly, Y, Z, T are interior points of edges BC, CD, DA , respectively. Prove:

- (a) If $\angle DAB + \angle BCD \neq \angle CDA + \angle ABC$, then among the polygonal paths, there is none of minimal length.
- (b) If $\angle DAB + \angle BCD = \angle CDA + \angle ABC$, then there are infinitely many shortest polygonal paths, their common length being $2AC \sin(\alpha/2)$, where $\alpha = \angle BAC + \angle CAD + \angle DAB$.

1971/5.

Prove that for every natural number m , there exists a finite set S of points in a plane with the following property: For every point A in S , there are exactly m points in S which are at unit distance from A .

1971/6.

Let $A = (a_{ij})$ ($i, j = 1, 2, \dots, n$) be a square matrix whose elements are non-negative integers. Suppose that whenever an element $a_{ij} = 0$, the sum of the elements in the i th row and the j th column is $\geq n$. Prove that the sum of all the elements of the matrix is $\geq n^2/2$.